

Chapter 2: Propagation & Frequency Bands

Syllabus Source: CAE Oxford (pp. 33-60) | Pass Target: >= 90% AviationExam

1. THE RADIO FREQUENCY SPECTRUM & AERONAUTICAL BANDS

Band Name	Frequency Range	Aeronautical Applications
VLF (Very Low Frequency)	3 to 30 kHz	Worldwide navigation, submarine communications.
LF (Low Frequency)	30 to 300 kHz	Non-Directional Beacons (NDB), commercial broadcast.
MF (Medium Frequency)	300 to 3,000 kHz	NDBs, commercial AM radio (subject to night effect).
HF (High Frequency)	3 to 30 MHz	Long-range oceanic & polar aeronautical communications (Skywave).
VHF (Very High Frequency)	30 to 300 MHz	Aeronautical RTF (118.000 - 136.975 MHz), VOR (108.0-117.95 MHz), ILS Localizer (108.1-111.95 MHz), Emergency (121.500 MHz).
UHF (Ultra High Frequency)	300 to 3,000 MHz	ILS Glidepath (329.15-335 MHz), DME, SSR (1030/1090 MHz), GPS (1575.42 MHz), Military Guard (243.0 MHz).
SHF (Super High Frequency)	3 to 30 GHz	Airborne Weather Radar (9 GHz), MLS (5 GHz), Satellite communications.

2. VHF PROPAGATION & THEORETICAL RANGE FORMULA

VHF radio waves travel essentially by direct 'space wave' (line of sight). Atmospheric refraction bends waves slightly downward around the curvature of the Earth (effective earth radius = 4/3 of true radius):

[!] EXAM TRAP (AVIATIONEXAM) - The Master VHF Radio Range Formula (AviationExam Absolute Core)

Range (NM) = 1.23 x (sqrt(h_aircraft [ft]) + sqrt(h_ground_station [ft])) -
Example 1: Aircraft at FL 100 (10,000 ft), ground station at sea level (0 ft): Range = 1.23 x sqrt(10,000) = 1.23 x 100 = 123 NM. - Example 2: Aircraft at FL 360 (36,000 ft), ground station at 400 ft: Range = 1.23 x (sqrt(36,000) + sqrt(400)) = 1.23 x (189.7 + 20) = 1.23 x 209.7 = ~258 NM! - Note: Optical horizon uses 1.06; Radio horizon uses 1.23 due to atmospheric refraction!

3. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)

[*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: VHF Line-of-Sight Range and 8.33 kHz Channeling

SCENARIO (AviationExam Standard Radiotelephony Calculation): An airliner is cruising at FL 360 (36,000 ft) en route to Madrid: - The Madrid ATC VHF transmitter antenna is located on a hill at 400 ft AMSL QUESTION: What is the maximum theoretical VHF two-way communication range in Nautical Miles (NM)? SOLUTION PASO A PASO (FOR DUMMIES): Step 1: Identify the Master Refracted Line-of-Sight Formula: - Master Equation: Range (NM) = $1.23 \times [\sqrt{h_{\text{aircraft}}} + \sqrt{h_{\text{station}}}]$ - (Factor is 1.23 for radio waves due to atmospheric refraction, vs 1.06 for visible light!). Step 2: Calculate the Square Roots of the Heights: - Aircraft Height $h_1 = 36,000 \text{ ft} \rightarrow \sqrt{36,000} = 189.74$ - Ground Station Height $h_2 = 400 \text{ ft} \rightarrow \sqrt{400} = 20.00$ - Sum of roots = $189.74 + 20.00 = 209.74$ Step 3: Multiply by the 1.23 Refraction Factor: - Range = $1.23 \times 209.74 = 257.98 \text{ NM}$ (~258 Nautical Miles)! FINAL ANSWER: Maximum VHF communication range is 258 NM! Beyond this distance, the direct VHF space wave is blocked by the curvature of the Earth.

4. HF SKYWAVE PROPAGATION & THE IONOSPHERE

High Frequency (HF: 3-30 MHz) propagates over thousands of miles via ionospheric refraction (skywave):

- Ionospheric Layers: D layer (50-90 km, daytime only, ABSORBS HF waves); E layer (90-140 km); F1 & F2 layers (140-400 km, merge into single F layer at night, REFLECTS HF waves back to Earth).
- Day vs Night Frequencies: DAY: Higher frequencies (10-18 MHz) required to penetrate the absorbing D layer and reach the F layer. NIGHT: Lower frequencies (3-8 MHz) required because the D layer vanishes and the F layer refracts lower frequencies.
- Dead Zone (Skip Zone): The area between the outer limit of the ground wave and the first returning sky wave where NO HF SIGNAL can be received.
- SELCAL (Selective Calling): An automated ground-to-air audio tone signaling system (4 letters, e.g. AB-CD). Alerts flight crew with chime and light, eliminating the need to listen to continuous static noise on HF.

4. 8.33 KHZ CHANNEL SPACING TRANSITION

To relieve severe VHF channel congestion in European airspace, the channel spacing was reduced from 25 kHz to 8.33 kHz (dividing each 25 kHz channel into three 8.33 kHz channels). Carriage of 8.33 kHz capable radio equipment is mandatory across all controlled airspace in the European ICAO region.

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)**[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Algoritmo Paso a Paso (Guía para Dummies): Number Pronunciation & Reading**

REGLAS MNEMOTÉCNICAS DE EXAMEN: 1. Pronunciación de números enteros: - Centenas redondas: 'HUND-red' (ej. 800 = 'AIT HUND-red'). - Millares redondos: 'TOU-SAND' (ej. 5.000 = 'FIFE TOU-SAND'). - Combinación: 38.300 = 'TREE AIT TOU-SAND TREE HUND-red'. 2. Niveles de vuelo (FL): SIEMPRE dígito a dígito (ej. FL 100 = 'FLIGHT LEVEL WUN ZE-RO ZE-RO'). 3. Escala de legibilidad: Del 1 (ininteligible) al 5 (perfecto). 'Read you 3' significa 'legible con dificultad'.

5. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: What is the theoretical VHF radio communication range between an aircraft at FL 250 (25,000 ft) and a ground station at 400 ft AMSL?

- [A]: 123 NM
- [B]: 195 NM
- [C]: 250 NM
- [D]: 320 NM

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [B]. Formula: $\text{Range} = 1.23 \times \sqrt{h} = 1.23 \times \sqrt{25,000} = 1.23 \times 158.11 = 194.5 \text{ NM}$ (~195 NM).

Q2: Why must a higher frequency band be selected for HF communications during the daytime compared to night?

- [A]: Atmospheric pressure is higher during the day
- [B]: The D layer of the ionosphere forms during daytime and strongly absorbs lower HF frequencies
- [C]: Solar wind turns off the F layer
- [D]: Transmitters require cooling at night

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [B]. Solar UV radiation ionizes the lower D layer during the day, which heavily absorbs low frequencies; higher frequencies are required to penetrate to the reflecting F layer.

Q3: What is the 'Dead Space' (Skip Zone) in HF radio communications?

- [A]: An area where aircraft engines fail
- [B]: The geographic area between the limit of the ground wave and the first returning sky wave where no signal is received
- [C]: The blind cone directly above a VOR station
- [D]: Space above the ionosphere

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [B]. The skip zone is the silent gap between the ground wave coverage limit and the touchdown point of the first ionospheric skywave.

Q4: What is the primary operational function of the SELCAL (Selective Calling) system?

- [A]: Automatically transmitting emergency MAYDAY messages
- [B]: Relieving the flight crew from continuously monitoring noisy HF audio frequencies
- [C]: Calculating aircraft ground speed
- [D]: Encoding digital transponder codes

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [B]. SELCAL monitors the HF frequency and alerts the crew with a chime and visual annunciator only when the ground station calls their specific 4-letter code.

Q5: In which frequency band does the international aeronautical emergency frequency 121.500 MHz operate?

- [A]: HF (High Frequency)
- [B]: VHF (Very High Frequency)
- [C]: UHF (Ultra High Frequency)
- [D]: LF (Low Frequency)

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CORRECT: [B]. 121.500 MHz is located in the VHF band (30 to 300 MHz).

Q6: What is the primary benefit of introducing 8.33 kHz channel spacing in European VHF communications?

- [A]: Doubling transmitter power
- [B]: Tripling the number of available communication channels to alleviate severe spectrum congestion
- [C]: Increasing VHF range to 1,000 NM
- [D]: Eliminating thunderstorm static

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [B]. Dividing 25 kHz channels into three 8.33 kHz channels triples available frequency channels.